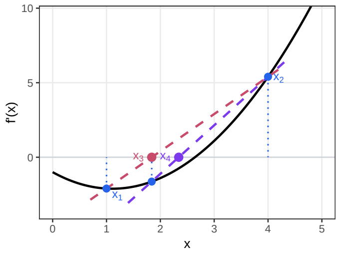
Optimization in Machine Learning

Univariate optimization
Further remarks



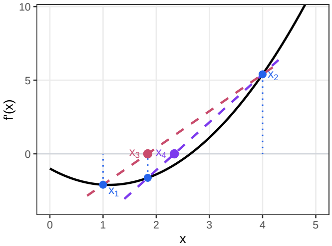
Learning goals

- Alternative methods
- Strategies for non-unimodal functions
- Software implementations

Slides based on ▸ Press et al. 1992 (Ch. 9 & 10)

ALTERNATIVE UNIVARIATE METHODS

- GSS, quadratic interpolation, and Brent are derivative-free
- Alternatively, if f' is available: find a minimum by solving $f'(x) = 0$ using root-finding methods applied to f':
  - **Bisection**: halve interval, keep the half where f' changes sign (slow, but guaranteed convergence)
  - **Secant method**: fit a secant through the interval bounds, use its intersection with x-axis as x_{new} (fast, but may diverge)



  - **Newton's method**: $x_{n+1} = x_n - f'(x_n)/f''(x_n)$ (quadratic convergence, but requires f'' and fails if $f'' \approx 0$)

WHAT IF F IS NOT UNIMODAL?

- Throughout this chapter, we have assumed that f is unimodal on the search interval
- Without unimodality, all covered methods may fail to find the global minimum:
  - **GSS** / **Quadratic Interpolation** / **Brent**: bracket shrinking discards the half with the larger function value, but the global minimum may lie in the discarded half
  - **Bisection** / **secant** / **Newton on** f': find any stationary point of f, which may be a local minimum, local maximum, or inflection point

DEALING WITH NON-UNIMODAL FUNCTIONS

- Practical heuristics:
  - multiple restarts: run the method from several different initial brackets and return the best result
  - e.g., split the domain into sub-intervals that are approximately unimodal and apply a local method to each

- Alternative: use global optimization methods (see next chapters), although finding the global minimum is still not guaranteed

SOFTWARE IMPLEMENTATIONS

- Several standard implementations for univariate optimization implement Brent’s method as default:
  - Python: `scipy.optimize.minimize_scalar()`
  - R-function `optimize()`
  - MATLAB: `fminbnd()`